

Bishop's University
Department of Computer Science
Data Visualization (CS503)
Winter 2024 – HW1

Part I: Dumbbell Chart

You have been given a dataset containing information on the traffic fatality rate per 100 million vehicle miles traveled in American united state. The dataset is stored in an Excel sheet named 'Dumbbell.xlsx.'

Using Python Notebook (Colab Jupyter or Anaconda Jupyter), your task is to reproduce (same shape, color, etc.) the dumbbell chart shown in the 'Dumbbell 1' tab of the Excel file.

You have the option to utilize the following libraries:

matplotlib (including matplotlib.pyplot) and plotly (including plotly.express, plotly.graph_objects), and Pandas and/or Seaborn libraries for data manipulation.

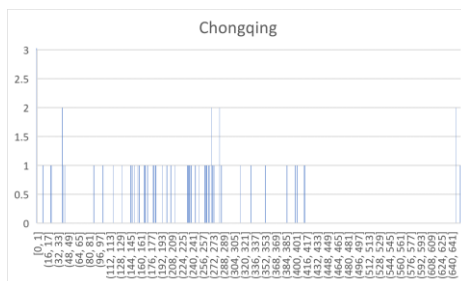
As additional assistance, you are provided with the following supporting documentation:

1. https://www.youtube.com/watch?v=qsOcLlo5vYc&ab_channel=CurbalDataLabs
2. <https://community.plotly.com/t/how-to-make-dumbbell-plots-in-plotly-python/47762>
3. <https://plotly.com/python/dumbbell-plots/>
4. https://github.com/iturki/Data-Analysis-and-Visualization-Projects/blob/master/dumbbell-chart-python/dumbbbell_plot.py
5. <https://medium.com/@ginoasuncion/creating-a-dumbbell-plot-with-plotly-python-570ff768ff7e>
6. <https://github.com/Shivank12batra/dumbbell-chart-plot/blob/master/Dumbell%20Charts.ipynb>
7. <https://www.kaggle.com/code/imoore/easy-tutorial-for-plotting-with-python/notebook>

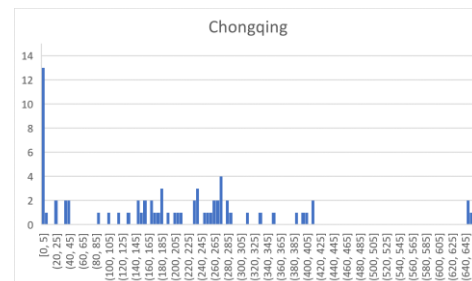
Part II: Histogram and Heatmap

A dataset about daily covid cases in China is provided in the file covid_china_daily_70days_2.csv.

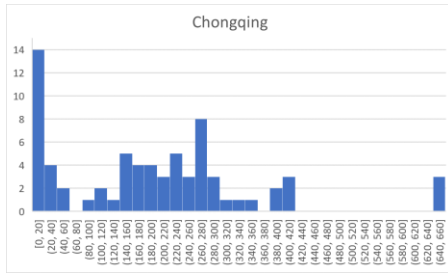
The histograms below illustrate the daily case distribution for the province of Chongqing, employing bin sizes of 1, 5, 15, 50, and 100.



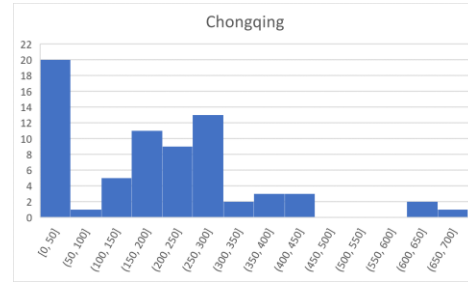
(a)



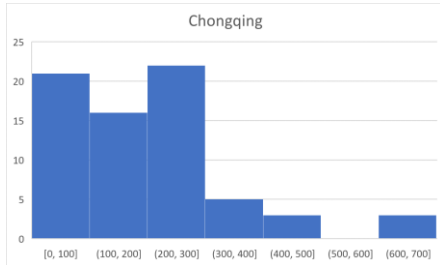
(b)



(c)



(d)



(e)

1. Using Python, reproduce the same histograms.
2. Calculate the distance between each pair of provinces' histograms using the Bhattacharyya Coefficient. Save the results in a square matrix of size NxN, where N is the number of provinces available. Refer to this matrix as the similarity matrix.
3. Plot the similarity matrix as a heatmap.
4. Redo 2 and 3 with different bin sizes of 1, 5, 15, 50, and 100.
5. Analyze and interpret the results by comparing the generated heatmaps.
6. Explore the impact of bin size on the visualization of heatmaps.

Bhattacharyya Coefficient

The Bhattacharyya coefficient is a measure of overlap between two statistical samples or populations. It quantifies the similarity between two probability distributions. Mathematically, the Bhattacharyya coefficient between two discrete probability distributions can be calculated as follows:

$$BC(P, Q) = \sum_i \sqrt{P(i)Q(i)}$$

P and Q represent two discrete distributions obtained by normalizing two distinct histograms. P(i) and Q(i) denote the probabilities associated with the ith bin.

To normalize a histogram, divide the height of each bin by the total number of values.

For more information about the Bhattacharyya coefficient, read the following tutorial:

<https://safjan.com/understanding-bhattacharyya-distance-and-coefficient-for-probability-distributions>